

Understanding Collaborative Practices in Distributed Agile Development

Research Proposal

Sunila Modi
Brunel University

School of Information Systems, Computing & Mathematics
Uxbridge, Middlesex, UK
sunila.modi@brunel.ac.uk

Pamela Abbott
Brunel University

School of Information Systems, Computing & Mathematics
Uxbridge, Middlesex, UK
pamela.abbott@brunel.ac.uk

Abstract— Agile methodologies in software development are gaining prominence in global settings where distributed cross-cultural software teams collaborate on projects, addressing the needs of dynamic business environments. However, undertaking collaborative work in distributed settings can create significant challenges, especially as a number of agile principles and practices rely on physical proximity and face-to-face interactions. An interpretative qualitative approach using case studies has been adopted to gain deeper insights into how such teams work. The analysis is being informed by theoretical concepts of awareness, common ground and boundary objects. Early findings suggest that distributed team members use multifaceted techniques, such as shared software artifacts that evolve over time to negotiate and enhance common ground for better collaborative practices to emerge. It is expected that research undertaken will provide a framework to understand this process. Practically, the research will propose guidelines for global software teams planning to adopt agile methods in distributed environments.

Keywords — *distributed collaborative work, globally distributed agile teams, boundary objects, establishing and negotiating common ground, awareness cues.*

I. INTRODUCTION

Agile methods have been in existence for over a decade and have brought significant changes to software engineering, in terms of values, principles and practices [11]. Whilst plan-driven and agile methods are intended to improve software development processes, agile methods require a distinctly different set of approaches. Traditional waterfall methods follow a sequential path, seeking stability in the project requirements and a predetermined plan of action. Whereas agile methods try to add value to the business by using an iterative and incremental approach where software developers work on small modules and respond to users' evolving business requirements. Hence agile methods try to provide the flexibility needed, as well as enough rigor to the software development process by accepting "feedback and change" and "embrace, rather than reject, higher rates of change" [28]. The agile values are fundamental and govern the agile software development process [15]. They are popular, because they provide flexibility and establish close communication with users. The agile methods were originally developed for collocated environments, as the tenets include: the need for physical proximity, reliance on face-to-face communication, cooperation on all parts of the project and a suitable

collaborative environment where trust and shared understandings can be developed through a team culture [15].

Recently, agile methodologies are increasingly gaining prominence in global settings where distributed software teams are set-up to work on projects, to address the need for accelerated deadlines of dynamic business environments; thus reducing expenditures by assigning work to countries with lower labor costs and increasing market proximity [2,22]. Adopting agile methods in globally distributed environments can create significant challenges especially as a number of agile principles and practices rely on physical proximity and face-to-face interactions.

II. RESEARCH FOCUS

The motivation of this research comes from exploring how collaborative practices take place within globally distributed cross-cultural agile teams and how distributed teams members accomplish common ground or a shared understanding to carry out collaborative work. With this aim in mind, we draw upon the theoretical lens of boundary objects, awareness cues and the theory of common ground to examine how they interrelate with each other to help explain how effective collaborative practices take place in globally distributed agile environments. We have identified the following research questions (RQ):

RQ 1: How do globally distributed agile development teams collaborate with each other to establish common ground or a shared understanding?

RQ 2: How do knowledge sharing interactions take place between distributed agile teams so that effective collaborative activities take place?

RQ 3: What transformations take place to the work practices in distributed agile teams so that collaborative work can be carried out in a continuous manner?

RQ 4: What can we learn from existing distributed agile development teams to convey to other global software teams who are planning to adopt agile methods in a distributed manner?

III. RELATED WORK

In this section, we explore the theoretical grounding for our study by unpacking the following concepts (a) common ground (b) boundary objects and (c) awareness cues.

A. Common Ground

One of the fundamental aspects of distributed agile teams is about communicative and coordinating actions in order to partake in collaborative work. Clark's theory of common ground regarding mutual knowledge, beliefs and assumptions maintains that the participants must have a shared awareness to carry out a joint activity [8]. It has been defined as "mutual knowledge, mutual beliefs, and mutual assumptions" between two parties where successful grounding in communication requires "to coordinate both the content and process" [8, 9]. In collocated environments establishing and negotiating common ground is a natural process, as the shared location provides the shared context whereby team members can be part of each other's workspace and share artifacts related to project easily. The theory of common ground originated in the linguistics and cognitive psychology but now has been applied in many other domains. The theory is especially useful when investigating distributed collaborative work where it is more challenging to achieve a mutually shared understanding. Clark & Brennan [8] use the term "grounding" to describe the process of developing mutually shared understanding where it allows actors to establish and negotiate common ground in an incremental manner. Failure to establish and maintain common ground can result in serious breakdowns in collaborative work [9].

B. Boundary Objects

The initial concept of boundary objects was introduced by Star & Griesemer [25] as "...objects which are both plastic enough to adapt to local needs and constraints of the several parties employing them, yet robust enough to maintain a common identity across sites" and they "sit in the middle of a group of actors with divergent viewpoints". They classify boundary objects into four functional categories: repositories, ideal type, coincident boundaries, and standardized forms. Where repositories can be objects such as a library or database. Ideal types are abstractions from different domains and may include diagrams, drawings, and clay models and are open to interpretation. Coincident boundaries are objects which have the same boundaries but different internal contents, for example an office building can a coincident boundary because representatives from different organizations can all work within the same physical boundary. Lastly, standardized forms are objects that provide different communities with a common method to communicate. From communities of practices (CoP) perspective Wenger [27] categorizes three types of boundary objects: (a) artifacts (tools, and documents, models shared by CoPs); (b) discourses: (a common language that can be shared across CoPs) and (c) processes: (shared processes, routines, procedures that facilitate coordination of and between CoPs).

The significance of the role of boundary objects has been increasing, especially in initiating and supporting collaborative work practices in distributed organizational contexts, where knowledge sharing across the boundaries is an essential part of

the work. Carlile [6,7] focuses on boundaries from a knowledge sharing perspective, in terms of syntactic, semantic and pragmatic boundaries. Where syntactic boundaries are the simplest, where knowledge can be transferred between actors providing that there is a common syntax. A semantic boundary is more complex because common meanings need to be developed in order to translate knowledge; for example, between a marketing department and a sales division they interpret what each other require in order to market and sell a product. Pragmatic boundaries are the most complex, as common interests need to be developed socially and politically to transform knowledge at a pragmatic boundary [7]. Boundary objects allow actors from different communities to reconcile their different meanings and produce shared understandings, which in turn enable cooperation between actors [3]. Furthermore, they provide a vehicle for convergence in meanings to establish shared understanding and through a process of transformation which enables creating common ground [3].

Within information systems development, examples of boundary objects are presented in the literature including requirements specification [1, 18] and developing prototypes [12]. Levina and Vaast [18] differentiate between boundary objects and "boundary objects-in-use", to examine some of the social and political dynamics of sharing knowledge across boundaries; where certain boundary objects become boundary objects-in-use when they are integrated meaningfully into the local practices and they maintain their common identity in joint practice. Barrett & Oborn [1] examine cross-cultural issues within the context of a global software development case study. They discuss how the role of boundary objects can have a dual function of enabling collaboration early on and impeding and hinder the collaborative process at later points in the software development process. Doolin & McLeod [12] draw on the sociomateriality dimension and boundary objects as a basis of analysis, where they trace how a prototype is developed and used within IS development. These studies build on the previous work on boundary objects [3,6] demonstrating the importance of studying boundary objects in facilitating knowledge sharing as well as the fragile and emergent nature they have in practice. In globally distributed agile development, the project artifacts such as the user stories, the shared code and test cases are an integral part of agile work practices; which could be seen as boundary objects in-use as they provide a common focus to the project goals and act mediators in the communication, coordination and cooperation processes between the team members across the boundaries.

C. Awareness cues

Drawing upon the literature of awareness within the computer-supported cooperative work (CSCW) field, awareness is defined as "an understanding of the activities of others, which provides a context for your own activities" [13]. Having appropriate awareness cues of team activities, tasks, artifacts and expertise of team members is essential in distributed software teams as it facilitates collaborative and cooperative activities to take place, which in turn supports the development of common ground. In distributed teams the awareness cues are reduced due to the distance therefore the members can face a lack of team and task awareness and

generally experience more challenges in maintaining awareness needed for the progress of tasks, the artifacts and their respective offshore team members [10, 17]. The 3C-collaboration model [14, 16] defines collaboration as “the union of communication, coordination and cooperation efforts”, where awareness is a contingent part of communication, coordination and cooperation processes. The model can be used as an evaluative mechanism, within an existing distributed agile development practices to gain an understanding of the on-going awareness challenges that distributed development teams face [20]. Furthermore the concepts of boundary objects, awareness cues and common ground can be interlinked (see Figure 1) [21]. Within the globally distributed agile settings, the project artifacts aid in raising awareness which subsequently contribute to establishing and negotiating common ground for joint collaborative practices to take place across the boundaries.

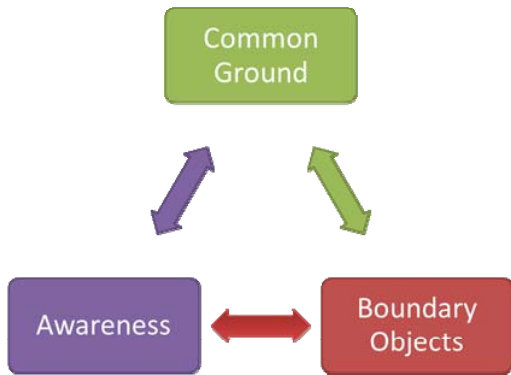


Figure 1: Interrelationship with common ground, awareness cues and boundary objects.

IV. RESEARCH METHODS

In order to address the research questions the following will be carried out (i) on-going review of the literature and (ii) to gain deeper insights of how distributed agile teams carry out collaborative practices an interpretative approach using case studies will be adopted. An interpretative approach is best suited where there is a complex relationship between people and processes, and where researchers are interested in a subjective understanding of the participants [26]. A case study approach allows an authentic representation of the situation to be explored [4, 29]. Case studies permit “thick descriptions” [29] which provide rich and valuable findings. A number of methods will be deployed for the data collection including (a) semi-structured interviews, (b) observations and (c) document analysis. The interview questions will be formulated beforehand; however, a large degree of flexibility will be incorporated to allow the researcher to pursue relevant issues which arise during the interviews and to allow open-ended answers from the interviewees. The interviews will be conducted with various team members of the distributed agile development teams. Consent from the participants will be gained to record the interviews. The data will be analyzed using thematic analysis from the literature sources [5]. The information will be coded into themes. This analysis will be combined with the field notes from the observations and the

document analysis to gain an overall sense of the insights from the case studies with relation to the review of the literature. Further comparisons of case studies will be carried out to identify the similarities and differences of the themes and patterns of how collaborative practices are supported in distributed agile settings.

V. RESEARCH PROGRESS

So far, we have examined one case study within an international bank and financial services company ABC Bank and its headquarters are based in the UK and it has distributed business centers and IT centers across the globe. The bank has set up captive centers in India where a number of teams work on an offshore basis. Our study is based on a distributed agile software development team (called the Operational Data Cache team), which is established across the London and India. The initial data collection of sixteen semi-structured interviews has been completed, where each interview lasted on average one hour. Eight face-to-face interviews with individuals from the London team and eight video-conference interviews with the India team counterparts were carried out. From examining the initial data, we reported on the communication challenges associated within distributed agile development [19]. Secondly the 3C Collaboration model was used for further analysis as an evaluative mechanism to explore how awareness was being raised across the borders; this was reported in [20]. Building on this line of research in order to address RQ1 and partially RQ2 the 3C collaboration model has been extended to understand how common ground or shared understanding is being established and negotiated within the distributed agile team [21].

VI. EXPECTED RESULTS

Our findings would provide us with a greater insight in the following (i) how distributed cross-cultural agile teams work in a collaborative manner and (ii) what transformations take place to the agile work practices and challenges the teams have to go through in order to carry out the collaborative work on an on-going basis. We therefore would contribute to the debate on how collaborative practices care out in globally distributed agile teams. Moreover, we expect to propose suggested guidelines for practitioners and senior management of global software teams of the key considerations they should focus on, when planning to adopt agile methods in geographically distributed environments.

VII. EVALUATION

The evidence of contribution from the study will be presented at suitable conferences and journals. We will be examining at least two case studies of global distributed agile teams in depth. Thus, the research would offer a platform for additional research which could go towards generalizability into how collaborative practices take place on an ongoing basis in globally distributed agile environments.

REFERENCES

- [1] M. Barrett and E. Oborn, “Boundary object use in cross-cultural software development teams,” *Human Relations*, vol. 63, no. 8, pp. 1199–1221, 2010.
- [2] D. Batra, “Modified agile practices for outsourced software projects,” *Commun. ACM*, vol. 52, (9), pp. 143–148, Sep. 2009.

- [3] B. A. Bechky, "Sharing meaning across occupational communities: The transformation of understanding on a production floor," *Organization Science*, vol. 14, (3), pp. 312–330, 2003.
- [4] I. Benbasat, D. K. Goldstein, and M. Mead, "The case research strategy in studies of information systems," *MIS quarterly*, pp. 369–386, 1987.
- [5] V. Braun and V. Clarke, "Using thematic analysis in psychology," *Qualitative research in psychology*, vol. 3, no. 2, pp. 77–101, 2006.
- [6] P. R. Carlile, "A pragmatic view of knowledge and boundaries: Boundary objects in new product development," *Organization science*, vol. 13, no. 4, pp. 442–455, 2002.
- [7] P. R. Carlile, "Transferring, translating, and transforming: An integrative framework for managing knowledge across boundaries," *Organization science*, vol. 15, no. 5, pp. 555–568, 2004.
- [8] H. H. Clark and S. E. Brennan, "Grounding in communication," *Perspectives on socially shared cognition*, vol. 13, no. 1991, pp. 127–149, 1991.
- [9] C. D. Cramton, "The Mutual Knowledge Problem and Its Consequences for Dispersed Collaboration," *Organization Science*, vol. 12, no. 3, pp. 346–371, May 2001.
- [10] D. Damian, L. Izquierdo, J. Singer, and I. Kwan, "Awareness in the wild: Why communication breakdowns occur," in *Global Software Engineering, 2007. ICGSE 2007. Second IEEE International Conference 2007*, pp. 81–90.
- [11] T. Dingsøyr, S. Nerur, V. Balijepally, and N. B. Moe, 'A decade of agile methodologies: Towards explaining agile software development', *Journal of Systems and Software*, vol. 85, no. 6, pp. 1213–1221, Jun. 2012.
- [12] B. Doolin and L. McLeod, "Sociomateriality and boundary objects in information systems development," *European Journal of Information Systems*, vol. 21, no. 5, pp. 570–586, 2012.
- [13] P. Dourish and V. Bellotti, "Awareness and coordination in shared workspaces," in *Proceedings of the 1992 ACM conference on Computer-supported cooperative work*, New York, NY, USA, 1992, pp. 107–114.
- [14] C. A. Ellis, S. J. Gibbs, and G. Rein, "Groupware: some issues and experiences," *Commun. ACM*, vol. 34, no. 1, pp. 39–58, Jan. 1991.
- [15] M. Fowler, and J. Highsmith "Manifesto for Agile Software Development." [Online]. Available: <http://agilemanifesto.org/>. [Accessed: 10-Feb-2013].
- [16] H. Fuks, A. Raposo, M. A. Gerosa, M. Pimentel, D. Filippo, and C. Lucena, "Inter- and intra-relationships between communication coordination and cooperation in the scope of the 3C Collaboration Model," in *12th International Conference on Computer Supported Cooperative Work in Design, 2008. CSCWD 2008, 2008*, pp. 148–153.
- [17] J. D. Herbsleb, "Global Software Engineering: The Future of Socio-technical Coordination," in *Future of Software Engineering, 2007. FOSE '07, 2007*, pp. 188–198.
- [18] N. Levina and E. Vaast, "The emergence of boundary spanning competence in practice: implications for implementation and use of information systems," *MIS quarterly*, pp. 335–363, 2005.
- [19] S. Modi, P. Abbott, and S. Counsell, "Exploring communication challenges associated with Agile practices in a globally distributed environment" *RAISE 2012- Researching Agile development of Information Systems Conference 11th Oct 2012 London*.
- [20] S. Modi, P. Abbott, and S. Counsell, "Raising Awareness in Distributed Agile Development : A Case Study Perspective" (2013). *UK Academy for Information Systems Conference Proceedings 2013*.
- [21] S. Modi, P. Abbott and S. Counsell, "Negotiating Common Ground in Distributed Agile Development: A Case Study Perspective" in *Global Software Engineering, 2013. ICGSE 2013. 8th IEEE International Conference, 2013*.
- [22] B. Ramesh, L. Cao, K. Mohan, and P. Xu, "Can distributed software development be agile?," *Commun. ACM*, vol. 49(10), pp. 41–46, Oct. 2006.
- [23] J. Sapsed and A. Salter, "Postcards from the edge: local communities, global programs and boundary objects," *Organization Studies*, vol. 25, no. 9, pp. 1515–1534, 2004.
- [24] S. L. Star, "This is not a boundary object: Reflections on the origin of a concept," *Science, Technology & Human Values*, vol. 35, no. 5, pp. 601–617, 2010.
- [25] S. L. Star and J. R. Griesemer, "Institutional ecology, translations' and boundary objects: Amateurs and professionals in Berkeley's Museum of Vertebrate Zoology, 1907–39," *Social studies of science*, vol. 19,(3), pp. 387–420, 1989.
- [26] G. Walsham, "Doing interpretive research," *European Journal of Information Systems*, vol. 15, no. 3, pp. 320–330, 2006.
- [27] E. Wenger, "Communities of practice and social learning systems," *Organization*, vol. 7, no. 2, pp. 225–246, 2000.
- [28] L. Williams and A. Cockburn, "Agile software development: it's about feedback and change," *Computer*, vol. 36, no. 6, pp. 39–43, Jun. 2003.
- [29] R. K. Yin, *Case Study Research: Design and Methods*. SAGE Publications, 2003.